REVIEW 2 | 23CSE452 BUSINESS ANALYTICS | ADVANCED ANALYTICS: TEXT MINING AND TIME SERIES

# Mobile App Update Impact and Failure Detection

What users complain about, what is coming next, and which app to fix first

1,137,987 Play Store reviews | 11 apps | 3 domains | 1 April to 20 September 2026


**Team number: 10**

Aditya Monish Kumar K - CB.SC.U4CSE23103

Regella Krishna Saketh - CB.SC.U4CSE23649

Akshay KS - CB.SC.U4CSE23104

Harshini Vennela - CB.SC.U4CSE23455

Kanishka D - CB.SC.U4CSE23155


**Live dashboard**
team10-app-reviews.streamlit.app

Review 2 presentation: 20 marks

# Three Business Questions, Two Unit 3 Methods

### 1 What are users complaining about?

Method 1, text mining: NMF topic models on 167,357 complaint reviews, VADER sentiment per topic, words that set each app apart.

### 2 How many complaints are coming, and was a release to blame?

Method 2, time series: ARIMA/SARIMA forecasts tested on a rolling backtest, unusual-day alerts, and before/after tests for 224 app versions.

### 3 Which app needs attention first, and what should it do?

Combined insights: six warning signs from every stage give each app a priority and 65 recommendations, all in a live dashboard.

## How the work flowed: one stage after another, each merged through a pull request

**R1**

**Review 1 data**

1.14M reviews, 9 issue tags, VADER, classifier

**1**

**Stage 1: preparation**

text corpora, daily series, dashboard base

**2**

**Stage 2: text mining**

17 topics, sentiment, evaluation

**3**

**Stage 3: forecasting**

11 ARIMA models, backtest

**4**

**Stage 4: evaluation**

accuracy, intervals, alerts

**5**

**Stage 5: impact + insights**

205 releases tested, priorities

# Text Mining: Preparing the Review Text for Topic Models

## Two corpora

**167,357**

complaint reviews (1–2★) kept, 75.5% of 221,581

**166,929**

praise reviews (4–5★) kept as a contrast, 19.2% of 871,380

**Kept:** at least 3 content words after cleaning. "worst" or "good" alone cannot carry a topic.

**Median length:** 11 words for complaints, 4 for praise. 3★ reviews are left out.

## Cleaning steps

1. Lowercase; remove links, e-mails, emojis, digits
2. Remove brand phrases ("google pay", "phone pe") so "pay" stays usable
3. Expand contractions: "didnt" → "did not"
4. Drop stop words but keep negations: "not delivered" keeps its meaning
5. Hinglish: "nahi", "nhi" → "not"; drop filler words ("hai", "mera")
6. Drop app names, so topics describe problems, not apps
7. Lemmatise (WordNet): "refunded" → "refund", "orders" → "order"

## Example

**Raw:** "PLEASE STOP CHEATING US BY CHARGING DELIVERY CHARGES FOR ORDERS ABOVE 100..."

**Clean:** stop cheat charge delivery charge order people number fraud

## Document-term matrices

| Matrix | Reviews | Terms |
|---|---|---|
| Complaints · Food | 82,603 | 17,448 |
| Complaints · Shopping | 71,413 | 18,665 |
| Complaints · Payments | 13,341 | 2,982 |
| Praise · Food | 74,647 | 6,875 |
| Praise · Shopping | 75,672 | 7,097 |
| Praise · Payments | 16,610 | 1,763 |

*Words and two-word phrases ("not deliver", "customer care") in at least 10 reviews and at most half of them.*

# Time Series: Daily Complaint Counts, Data Gaps and Releases

### 1 The series

Daily count of 1–2★ reviews per app: 11 series of 173 days. Counts, not shares: the late-April feed gap removed mostly positive reviews, which distorts shares far more than counts.

### 2 Data-quality flags

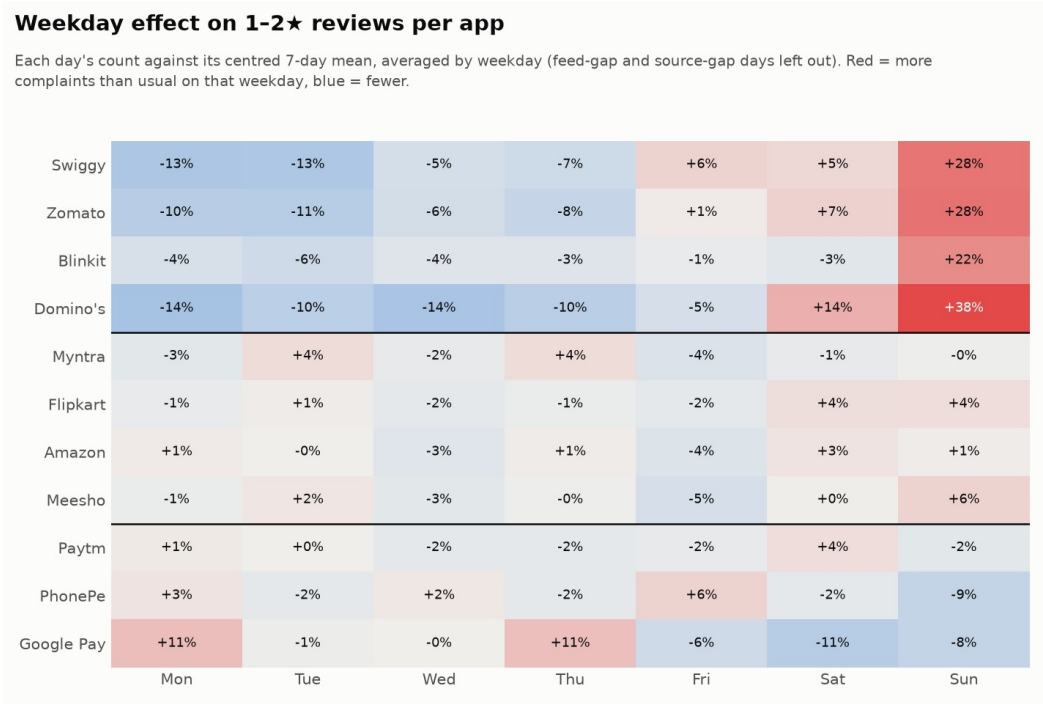
Feed gap 21 Apr – 5 May for five apps becomes a 0/1 input to ARIMA. Zomato's missing day (24 Jul) is filled with the same weekday one and two weeks either side.

### 3 224 release events

From 2,507 app versions: at least 30 reviews, first seen after 1 April, and numbered higher than every earlier version (old builds removed).

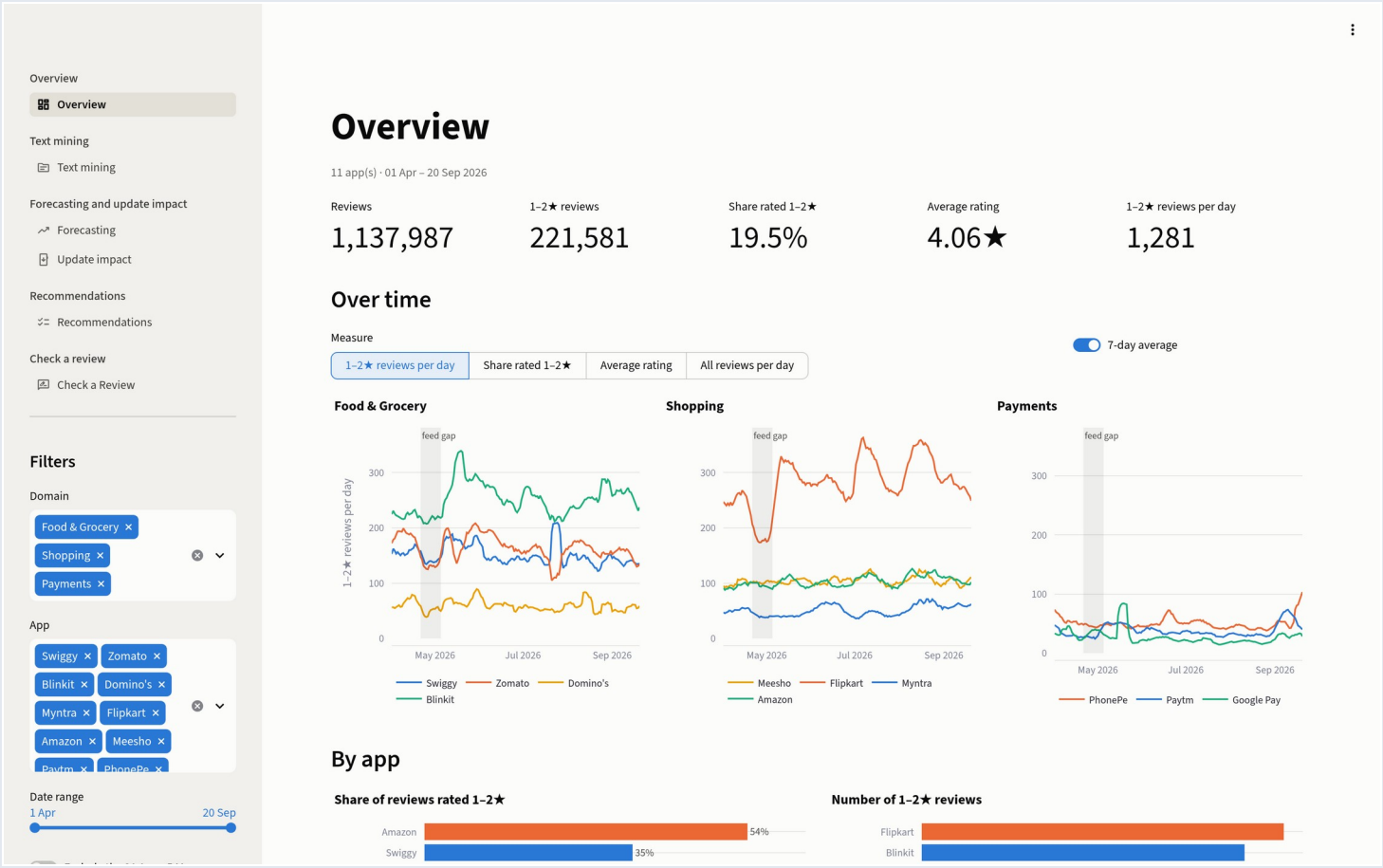
### 4 Checks for ARIMA

ADF test: 9 of 11 apps stationary in level, so d = 1 only for Zomato and Blinkit. STL seasonal strength below 0.64 everywhere, so D = 0.




**Weekly pattern:** food apps get 22–38% more complaints on Sundays; shopping apps have no weekday pattern. This is why only food apps get seasonal ARIMA terms.

# Dashboard: One App, Six Pages, Live on Streamlit Cloud




## Built in Stage 1

- Shared filters on every page: domain, app, date range, exclude the feed gap
- Reads small precomputed tables, so it opens in seconds
- Overview: volume, 1–2★ share, trends per domain, top issues

## Six pages

| Page | Built by |
|---|---|
| Overview | Aditya |
| Text mining | Akshay |
| Check a review | Harshini |
| Forecasting | Kanishka |
| Update impact | Kanishka |
| Recommendations | Kanishka |

[team10-app-reviews.streamlit.app](team10-app-reviews.streamlit.app)

*Redeploys from GitHub after every merge.*

# Topic Modelling: 17 Complaint Topics with NMF

## How the topics were found

**TF-IDF + NMF:** the counts become TF-IDF weights; NMF splits them into topics (V ≈ W × H), each a weighted list of terms.

**Why NMF:** on very short reviews LDA mixes themes; NMF with nndsvda start is deterministic and runs in seconds on 167,357 reviews.

**One model per domain:** K = 6 Food & Grocery, 6 Shopping, 5 Payments (chosen by coherence, next slide).

**Per review:** its dominant topic plus VADER sentiment, so every topic has a mean score and a share of strongly negative reviews (≤ −0.5).

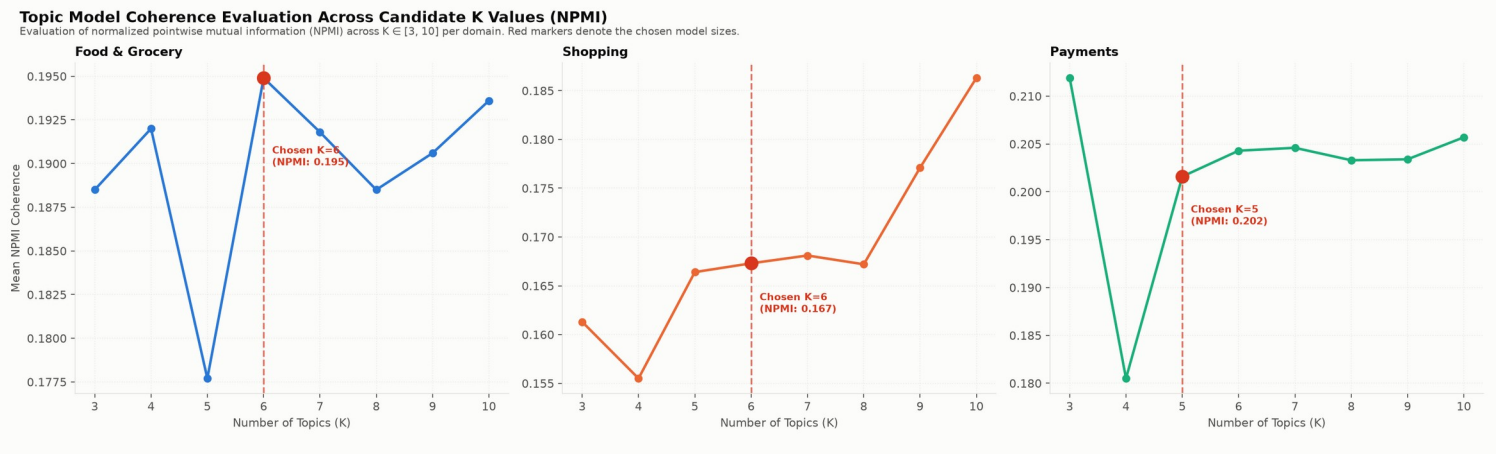
**Contrast:** a second NMF (K = 4 per domain) on 4–5★ reviews shows what satisfied users praise.

**Reused:** each pipeline is saved, so the dashboard can give a new review its topic.

| Domain | Complaint topic | Share | Mean VADER | Strongly neg. |
|---|---|---|---|---|
| **Food & Grocery** | Order Cancellation & Refund | 35% | -0.27 | 36% |
| | Customer Support & Bot Loop | 19% | **-0.40** | 56% |
| | Excessive Delivery & Handling Charges | 11% | -0.21 | 24% |
| | Poor Food Quality & Stale Items | 13% | **-0.44** | 59% |
| | Delivery Delays & Rider Tracking | 18% | -0.29 | 39% |
| | COD & Payment Option Failures | 5% | -0.12 | 18% |
| **Shopping** | Return Rejection & Defective Products | 25% | -0.23 | 33% |
| | Customer Service & Escalation Failure | 18% | -0.24 | 41% |
| | Courier Delays & Delivery Agent Issues | 16% | -0.29 | 41% |
| | Poor Shopping Experience & Fabric Quality | 9% | **-0.59** | 81% |
| | Extreme Dissatisfaction & Scam Allegations | 10% | **-0.61** | 80% |
| | Order Cancellation & Rescheduling | 22% | -0.31 | 41% |
| **Payments** | Failed Transfers & Debited Amounts | 52% | -0.15 | 27% |
| | App Crashes & Update Regressions | 9% | -0.09 | 16% |
| | Device Environment & Security Scan Errors | 10% | -0.11 | 18% |
| | Account Blocking & AutoPay Issues | 10% | -0.13 | 21% |
| | Customer Support Unresponsiveness | 18% | -0.23 | 40% |

*Share = share of the domain's topic-assigned complaints. Red = mean sentiment of −0.4 or lower.*

# Evaluation: Coherence, Keyword Tags and the Untagged 43%




## Choosing K (NPMI coherence)

K = 3 to 10 tried for each domain.

Food & Grocery peaks at K = 6 (0.195).

Shopping and Payments are almost flat, so we took the K where every topic is distinct and can be named; larger K split delivery and support into near-duplicates.

## 1 Topics agree with the keyword tags

The customer-support topics carry the Customer Support tag in 71% (food) and 76% (shopping) of their reviews.

Delivery-delay topic: 51% Delivery Delay tag. Device-environment errors: 46% Crash & Stability.

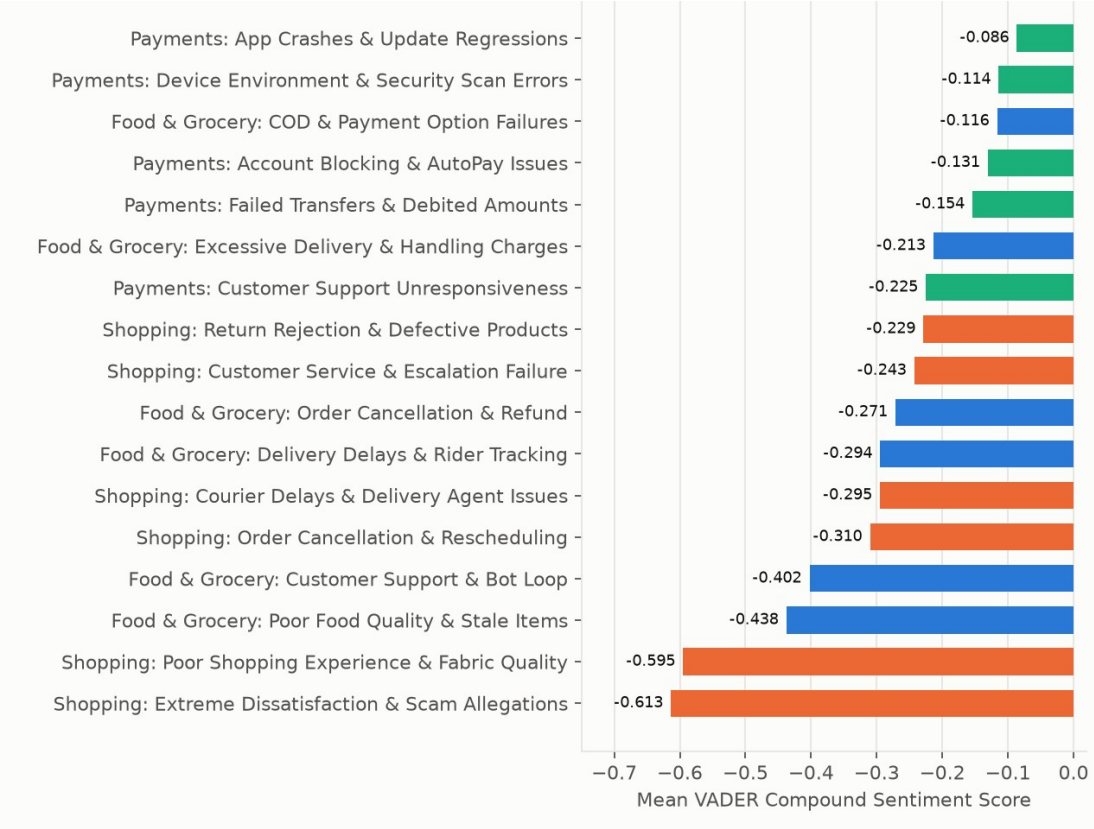
## 2 They also find what keyword rules miss

**71,929 of 167,357** complaints (43%) match none of the 9 keyword tags. The topics place each one: order cancellation and refund 13,073, returns 8,275, food quality 6,089.

Untagged share is highest for COD failures (80%) and failed transfers (72%): Hinglish and informal wording.

# Interpretation: The Angriest Complaints and How Apps Differ

### Mean VADER sentiment of each complaint topic




### Loss and broken trust are angriest

Shopping scam allegations −0.61 and poor quality −0.60 (about 80% strongly negative). Stale food (−0.44) angers more than late food (−0.29).

### Technical faults are reported calmly

Payment app crashes −0.09, device-environment errors −0.11: users describe them factually, so tone is not a measure of severity.

### Each app has its own complaint words

Swiggy: cancellation fee. Zomato: Gold, platform fee. Blinkit: damaged items. Myntra: exchange, size. Flipkart: Minutes, reschedule. Amazon: Prime membership. Meesho: account verification. Paytm: "device environment". PhonePe: charges. Google Pay: rewards.

### What satisfied users praise

Fast delivery (food), product quality and price (shopping), easy UPI payments.

# ARIMA Forecasting with a Rolling-Origin Backtest

## Three models, refitted at every step

**ARIMA / SARIMA (main):** weekly seasonal terms for the four food apps; the feed-gap flag as an extra input for five apps (SARIMAX); 80% and 95% intervals.

**Seasonal naive (baseline):** same weekday last week.

**Holt-Winters:** damped trend + weekly season.

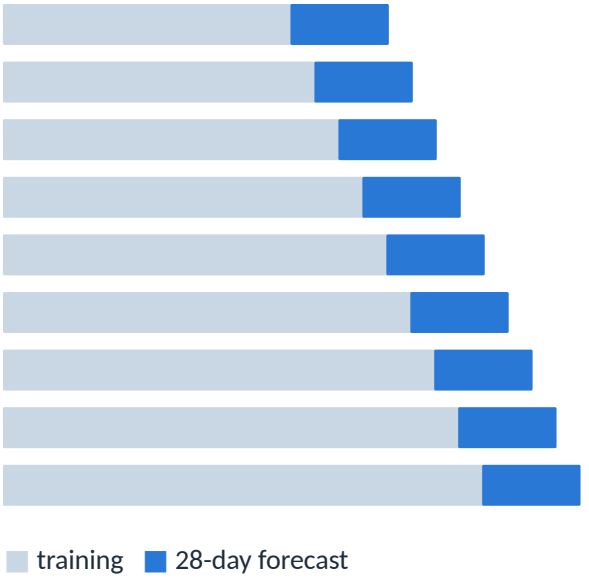
## Choosing the orders

- d from the ADF test; D = 0
- ACF/PACF propose p, q ≤ 3 and lag-7 seasonal terms
- Lowest AICc wins (378 fits); within 2 points, the simpler model
- No look-ahead: backtest orders use the first 84 days only

| App | Final order | Gap input |
|---|---|---|
| Swiggy | (0,0,1)(1,0,1) | yes |
| Zomato | (2,1,1)(1,0,1) | |
| Blinkit | (1,1,1)(1,0,1) | yes |
| Domino's | (0,0,1)(1,0,1) | yes |
| Myntra | (3,0,2) | |
| Flipkart | (1,0,1) | yes |
| Amazon | (1,0,1) | yes |
| Meesho | (0,0,2) | |
| Paytm | (2,0,2) | |
| PhonePe | (1,0,0) | |
| Google Pay | (1,0,0) | |

*(p,d,q)(P,D,Q) with a 7-day season. Only the food apps need seasonal terms, matching their Sunday peak.*

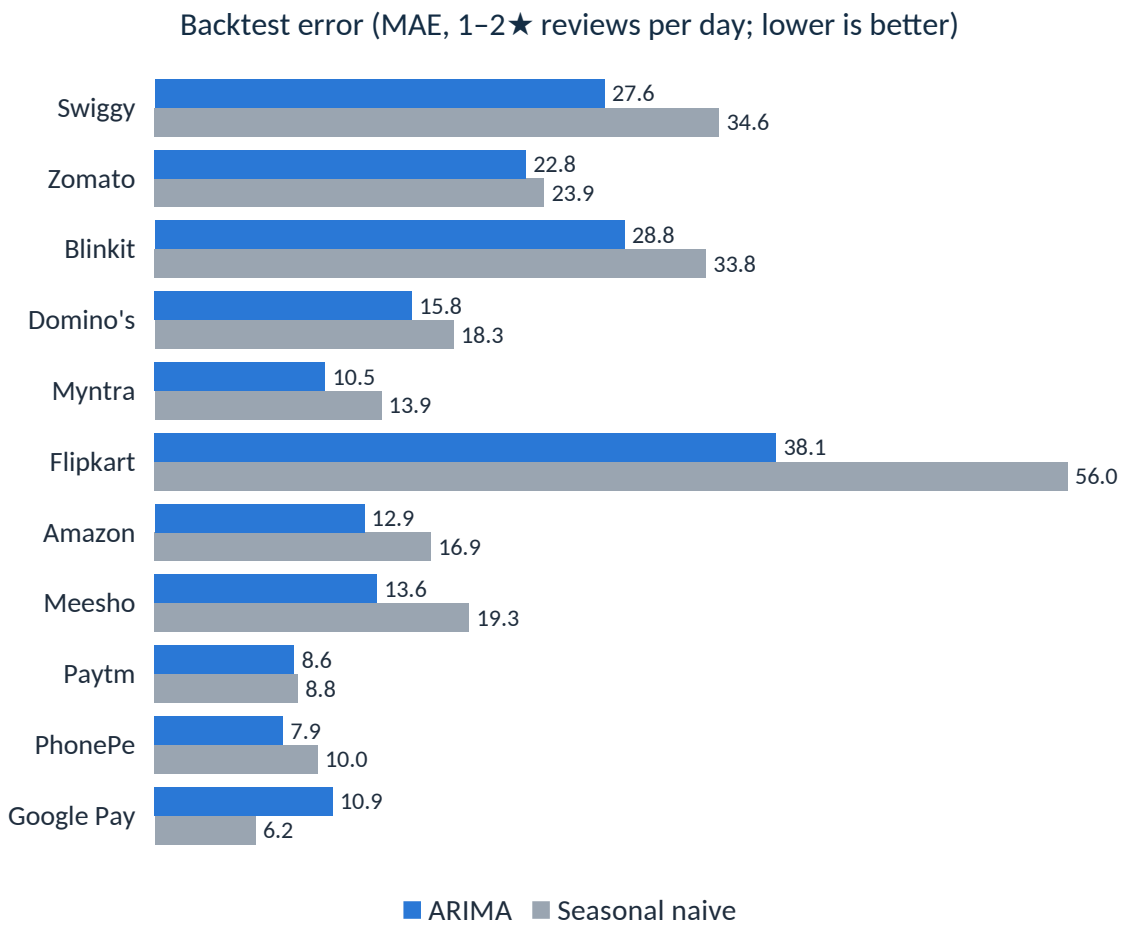
## Rolling-origin backtest




First 84 days to train, then forecast 28 days; move one week and refit.

**9 origins (23 Jun – 18 Aug) = 252 forecasts** per app and model.

# ARIMA Beats the Baseline for 10 of 11 Apps




**10 / 11**
apps where ARIMA has a lower error than the baseline

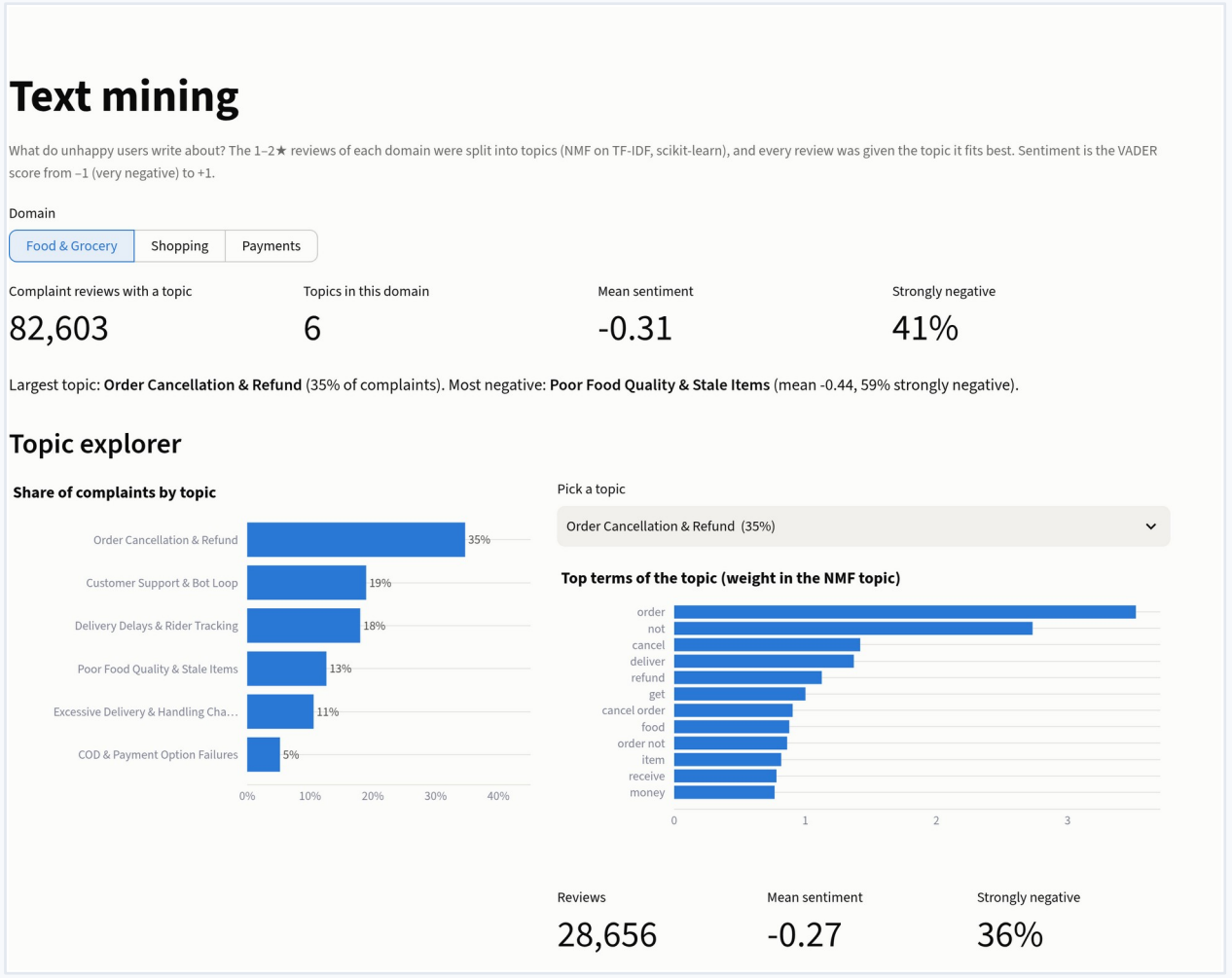
**−32%**
Flipkart error vs the baseline, the largest gain

**1 / 11**
Google Pay: the baseline wins (6.2 vs 10.9)

### Forecast for 21–27 Sep (1–2★ reviews, week 1)

| App | Forecast | 80% interval | vs last 4 wks |
|---|---|---|---|
| Flipkart | 1,830 | 1,596 – 2,056 | −7% |
| Blinkit | 1,711 | 1,488 – 1,929 | −8% |
| Swiggy | 1,037 | 889 – 1,180 | +7% |
| Zomato | 977 | 811 – 1,151 | −7% |
| Meesho | 737 | 676 – 801 | +4% |
| Amazon | 714 | 642 – 786 | −6% |
| Domino's | 413 | 327 – 495 | +7% |
| Myntra | 371 | 318 – 424 | −10% |
| PhonePe | 348 | 270 – 431 | −10% |
| Google Pay | 192 | 105 – 310 | +3% |
| Paytm | 136 | 75 – 205 | **−61% (unreliable)** |

# Dashboard: Text-Mining Page




## What a manager can do here

1. **Topic explorer:** pick a domain and a topic: its share, top terms, sentiment and real example reviews (189 readable examples, 3 per topic and app)
2. **Keyword-tag overlap:** which keyword tags the topic's reviews carry, and how many carry none
3. **Topics by app:** heatmap of each topic's share, or its share of strongly negative reviews
4. **Topic share over time:** monthly trend, following the date filter
5. **Distinctive terms:** the words that set one app apart from its domain

# Forecast Evaluation: Accuracy, Intervals and Residuals

## Accuracy over the backtest (mean of 11 apps)

| Model | MAE | RMSE | MASE | Beats naive |
|---|---|---|---|---|
| **ARIMA** | **17.95** | **24.95** | **0.88** | **10 of 11 apps** |
| Holt-Winters | 19.14 | 26.66 | 0.93 | – |
| Seasonal naive | 21.98 | 31.07 | 1.08 | baseline |

*MASE below 1 = more accurate than a seasonal-naive forecast on the training data.*

**81% / 93%**

of 2,772 forecasts fell inside the 80% / 95% intervals: close to nominal, 95% slightly narrow

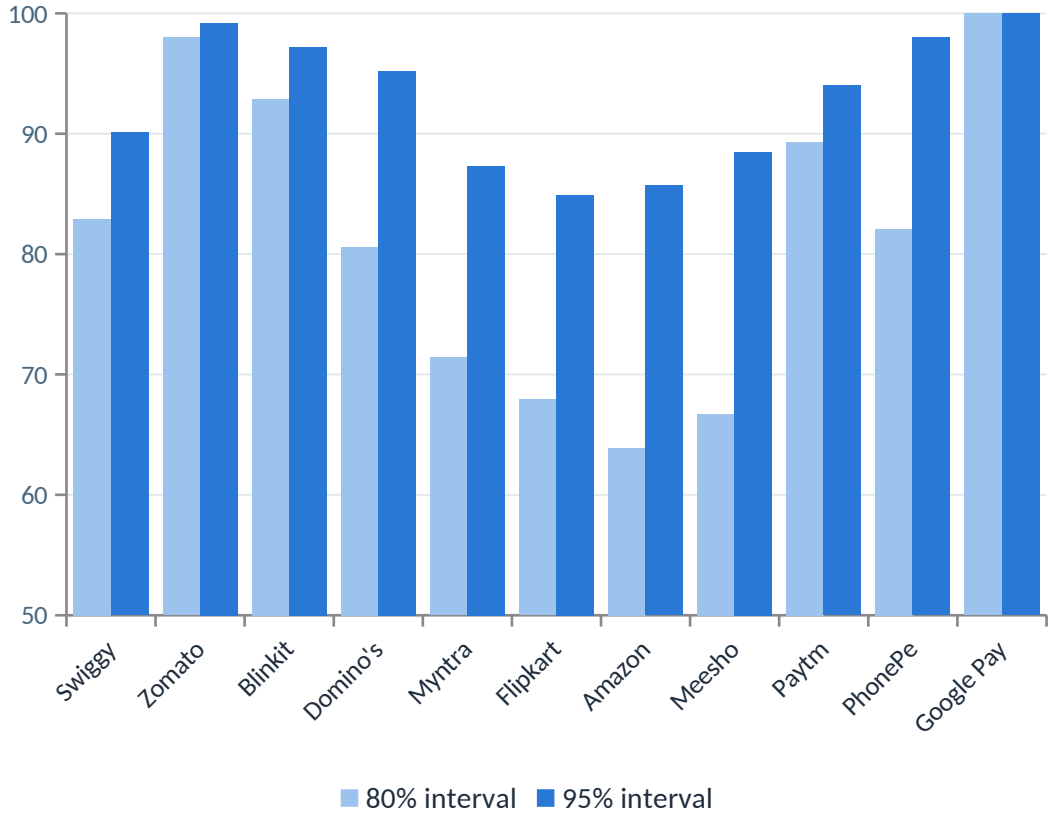
Rolling windows overlap, so this is a descriptive check, not an independent test.

**11 / 11**

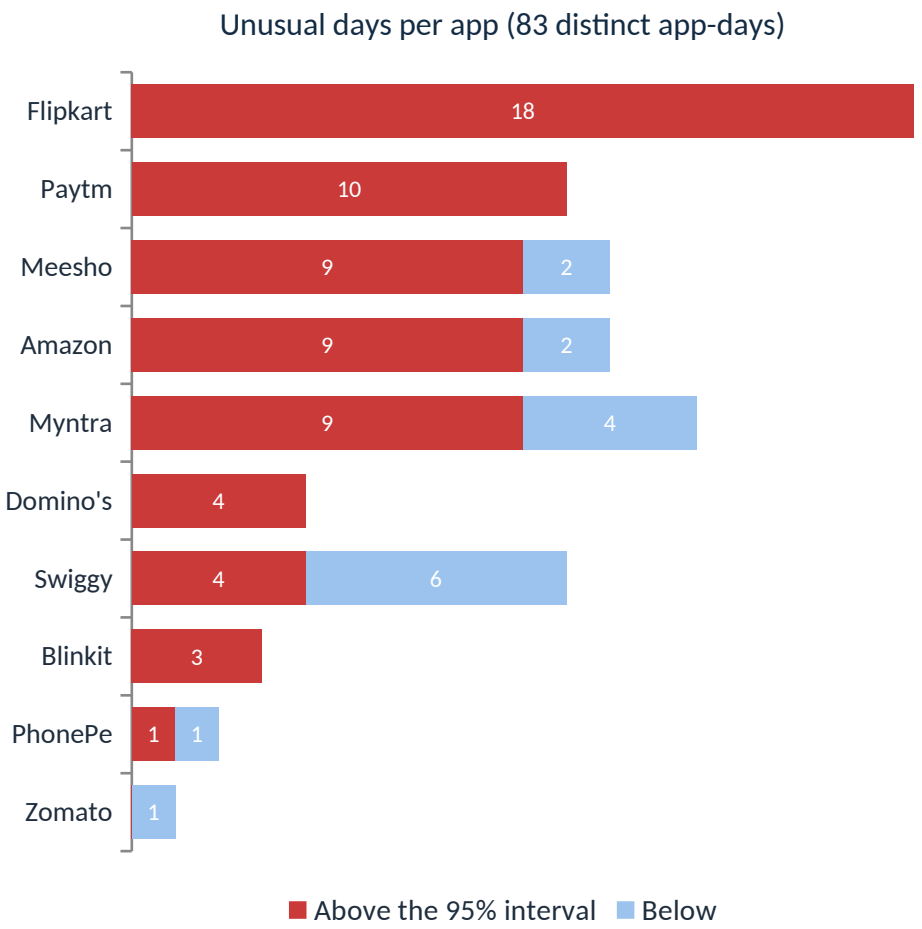
apps pass the Ljung–Box test at lag 14 ($p > 0.05$): no autocorrelation left in the residuals

Passing does not prove the residuals are random; it finds no pattern at that lag.

Share of backtest days inside the ARIMA interval (%); nominal 80 and 95



# Unusual Days, and Which Apps Can Be Predicted




### Alerts

A day is flagged when its count falls outside the 95% interval: 201 flagged forecasts = 83 distinct days (67 above, 16 below). Largest: Swiggy 24 Jul, 469 vs 157 expected; Flipkart 4 Jul, 451 vs 264.

### Predictable: the food apps

A steady weekly rhythm. Zomato's and Blinkit's 95% intervals hold 99% and 97% of days.

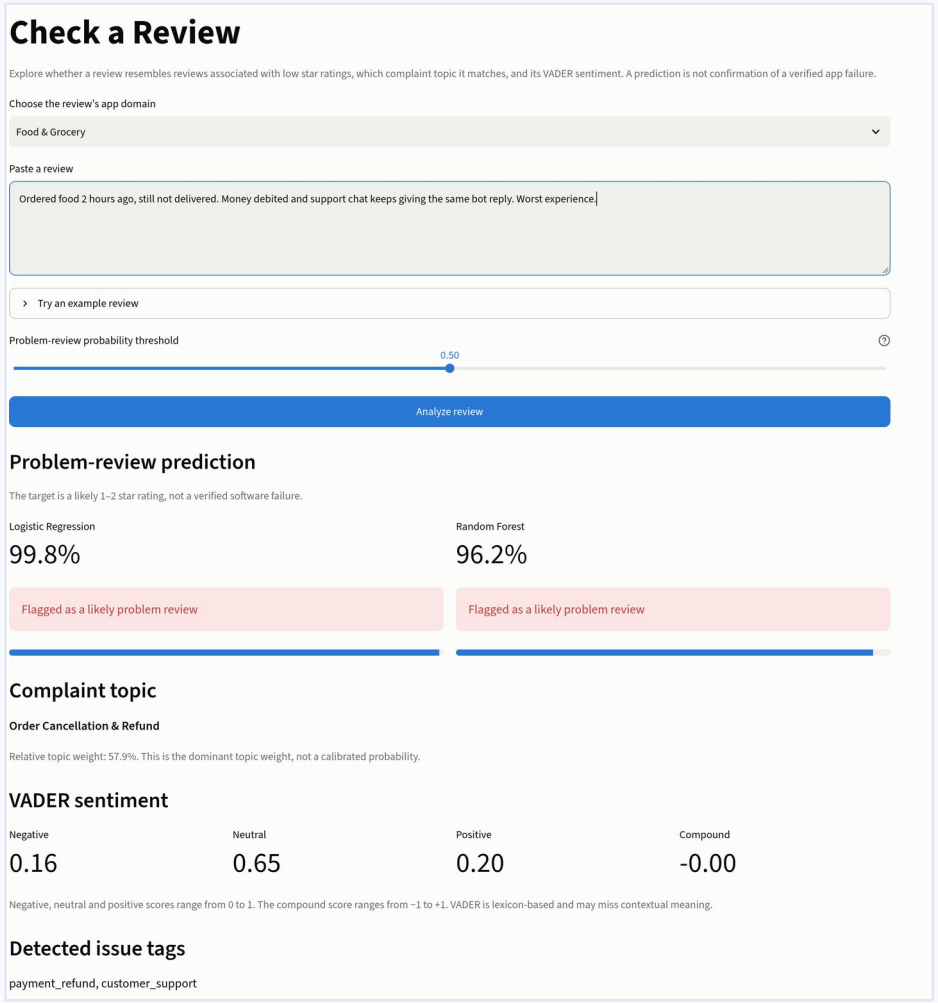
### Hard: shopping apps with level shifts

Flipkart, Amazon, Meesho, Myntra: the 80% interval holds only 64–71% of days, and they have the most alerts.

### Use with care: Google Pay and Paytm

Google Pay is low and noisy (the baseline wins). Paytm jumped to about 80 a day in September, so its week-1 forecast (−61%) is not reliable.

# Dashboard: Check a Review




## Paste any review and get four answers

1. **Problem review:** probability from the Review 1 Logistic Regression and Random Forest, with an adjustable threshold
2. **Complaint topic:** from the saved NMF pipeline of the chosen domain
3. **Sentiment:** VADER negative, neutral, positive and compound scores
4. **Issue tags:** the Review 1 keyword tags it matches

## In the example shown

"Ordered food 2 hours ago, still not delivered. Money debited..." is flagged by both models (99.8% and 96.2%), gets Order Cancellation & Refund, and the Payment & Refund and Customer Support tags.

Its VADER compound is 0.00: sentiment alone would miss this complaint, which is why the page combines all four.

# Update Impact: Did a New Version Make Things Worse?

## Two comparisons per release (7-day windows)

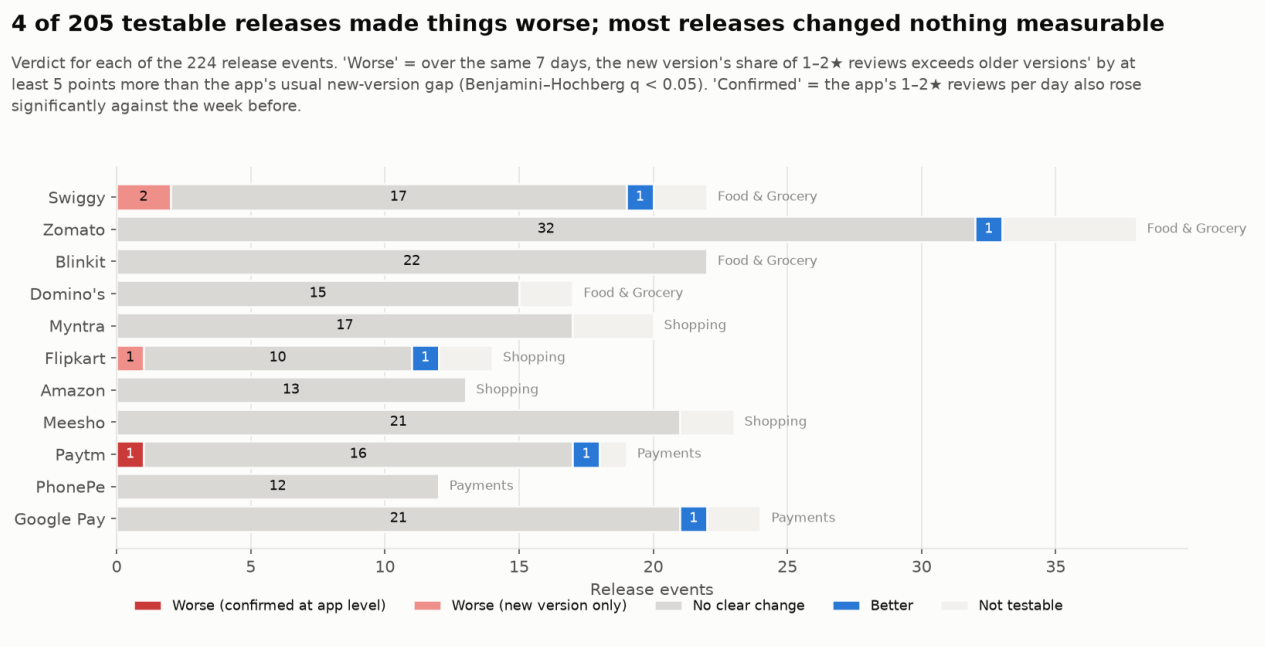
**A. Before vs after:** the app's 1–2★ reviews per day in the week before vs the week from the release (Welch t-test).

**B. New vs older version (main):** in the same 7 days, the 1–2★ share of users on the new version vs users still on older versions (two-proportion z-test). Same days, so sales and outages hit both groups.

**Usual gap:** new-version users on Amazon and Flipkart are always about 6 points more negative, so each release is compared with its app's usual gap.

**Worse:** at least 5 points beyond the usual gap, Benjamini–Hochberg q < 0.05 over all releases. Confirmed if A also rises.

**Release day:** the day the version got its 10th review; at least 30 reviews on each side, else not testable.




**224**
release events

**205**
testable

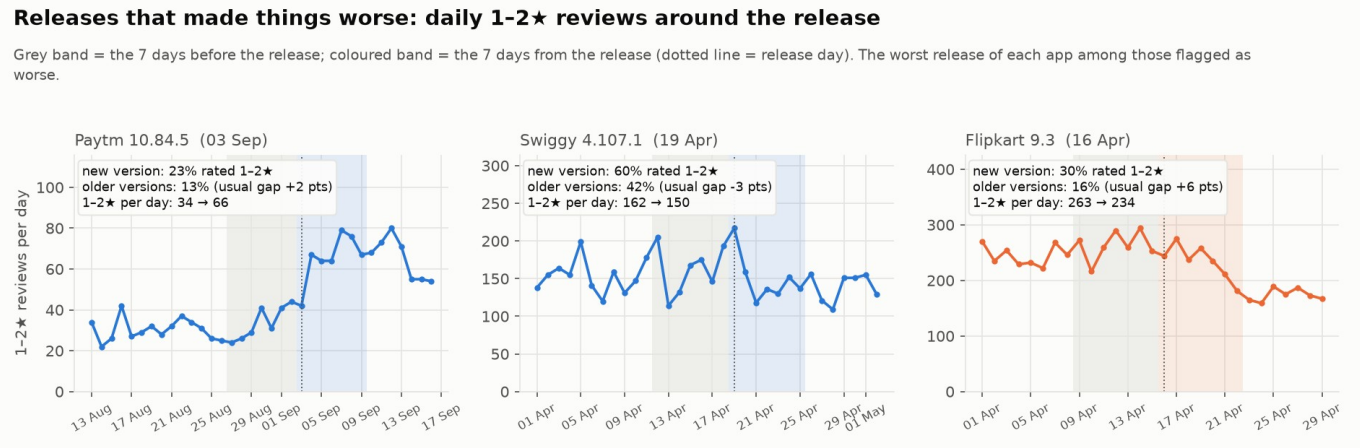
**196**
no clear change (96%)

**5**
better

**4**
worse, 1 confirmed

# Four Releases Made Things Worse; Paytm 10.84.5 Is Confirmed

| App | Version | Release | 1–2★ new vs older | Beyond usual gap (pts) | 1–2★ per day before → after | Issue that rose most | Verdict |
|---|---|---|---|---|---|---|---|
| **Swiggy** | 4.107.1 | 19 Apr | 60% vs 42% | +21.2 | 162 → 150 | Customer Support +8.9 | New version only |
| **Swiggy** | 4.107.2 | 23 Apr | 88% vs 83% | +8.5 | 163 → 137 | – | New version only |
| **Flipkart** | 9.3 | 16 Apr | 30% vs 16% | +7.6 | 263 → 234 | Customer Support +4.2 | New version only |
| **Paytm** | 10.84.5 | 3 Sep | 23% vs 13% | +8.1 | 34 → 66 | – | **Confirmed at app level** |




## How to read it

**Paytm 10.84.5 is the clearest bad release:** app-wide 1–2★ reviews nearly doubled (34 → 66 a day) and Harshini's forecast raised 6 alerts that week. It explains Paytm's September jump.

The other three have windows that overlap the late-April feed gap, so their shares are inflated.

No release made things worse for the other 8 apps. Same 4 flagged when the release day is the first review.

# Combining Every Stage: Which App Needs Attention First

## Six warning signs, one point each

1. **Many 1–2★ per day:** above the median app (103) time series
2. **High 1–2★ share:** above the all-app share (18.9%) time series
3. **Rising trend:** last 4 weeks +10% vs first 4 time series
4. **Forecast above recent:** reliable week-1 forecast +5% forecasting
5. **Days above forecast:** 5+ days above the 95% interval evaluation
6. **Release made it worse:** at least one worse release update impact

**Score:** 3+ High, 2 Medium, 0–1 Low. Every point traces back to one number from one stage.

**Which apps need attention first: six warning signs from all stages**

Filled dot = the sign applies. Score = number of signs (3+ High, 2 Medium, 0–1 Low). Signs: 1–2★ reviews per day above the median app; share of 1–2★ reviews above all apps; last 4 weeks 10%+ above the first 4; reliable week-1 forecast 5%+ above the last 4 weeks; 5+ days above the 95% forecast interval; a release that made things worse.

| | Many 1–2★ per day | High 1–2★ share | Rising trend | Forecast above recent | Days above forecast | Release made things worse | Score | Largest complaint topic |
|---|---|---|---|---|---|---|---|---|
| Swiggy · Food & Grocery | ● | ● | ○ | ● | ○ | ● | 4 High | Order Cancellation & Refund (37%) |
| Flipkart · Shopping | ● | ○ | ○ | ○ | ● | ● | 3 High | Return Rejection & Defective Products (25%) |
| Amazon · Shopping | ○ | ● | ● | ○ | ● | ○ | 3 High | Customer Service & Escalation Failure (24%) |
| Paytm · Payments | ○ | ○ | ● | ○ | ● | ● | 3 High | Failed Transfers & Debited Amounts (44%) |
| Blinkit · Food & Grocery | ● | ○ | ● | ○ | ○ | ○ | 2 Medium | Order Cancellation & Refund (33%) |
| Meesho · Shopping | ● | ○ | ○ | ○ | ● | ○ | 2 Medium | Return Rejection & Defective Products (28%) |
| Domino's · Food & Grocery | ○ | ● | ○ | ● | ○ | ○ | 2 Medium | Order Cancellation & Refund (41%) |
| Myntra · Shopping | ○ | ○ | ● | ○ | ● | ○ | 2 Medium | Return Rejection & Defective Products (26%) |
| Zomato · Food & Grocery | ● | ○ | ○ | ○ | ○ | ○ | 1 Low | Order Cancellation & Refund (33%) |
| Google Pay · Payments | ○ | ● | ○ | ○ | ○ | ○ | 1 Low | Failed Transfers & Debited Amounts (59%) |
| PhonePe · Payments | ○ | ○ | ○ | ○ | ○ | ○ | 0 Low | Failed Transfers & Debited Amounts (54%) |

**High:** Swiggy (4), Flipkart, Amazon, Paytm (3). **Medium:** Blinkit, Meesho, Domino's, Myntra. **Low:** Zomato, Google Pay, PhonePe.

# Recommendations, Each Backed by a Number

## Food & Grocery

**Fix first:** Fix the refund and cancellation flow: 35% of complaints (28,656), in every food app.

**Most damaging:** Food quality is the angriest topic (−0.44, 59% strongly negative): act fast on stale or wrong food.

## Shopping

**Fix first:** Fix return rejections and defective products: 25% of complaints, largest in Flipkart, Meesho and Myntra.

**Most damaging:** Answer scam allegations quickly and visibly: the most hostile reviews (−0.61).

## Payments

**Fix first:** Resolve debited-but-not-received payments: 52% of complaints.

**Most damaging:** Support that does not answer drives the anger (−0.23), more than the failure itself (−0.15).

### Swiggy · High (4)

- Fix refunds first (37% of its complaints)
- Review 4.107.1: support complaints +8.9 pts
- Plan for ~1,037 1–2★ reviews next week

### Flipkart · High (3)

- Fix return rejections (25%)
- Daily alert: 18 days above the forecast, the most
- Review 9.3: support and returns rose

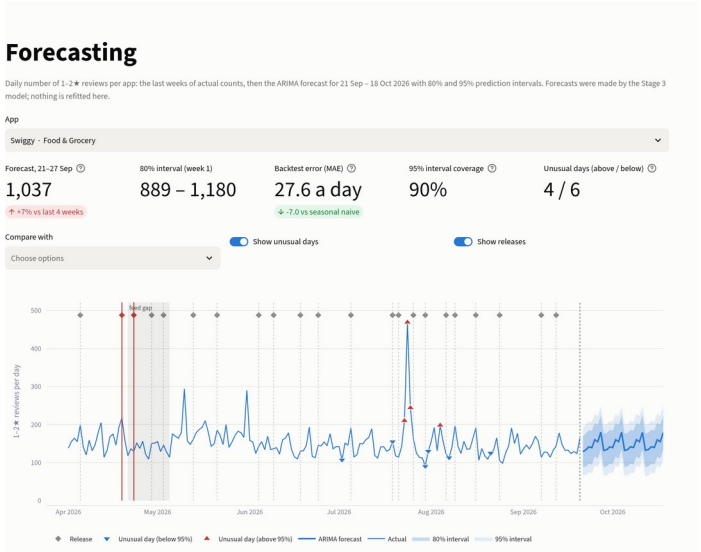
### Amazon · High (3)

- Fix escalations in customer service (24%)
- 53% of its reviews are 1–2★, the highest share
- Classifier triage works best here (91% precision)

### Paytm · High (3)

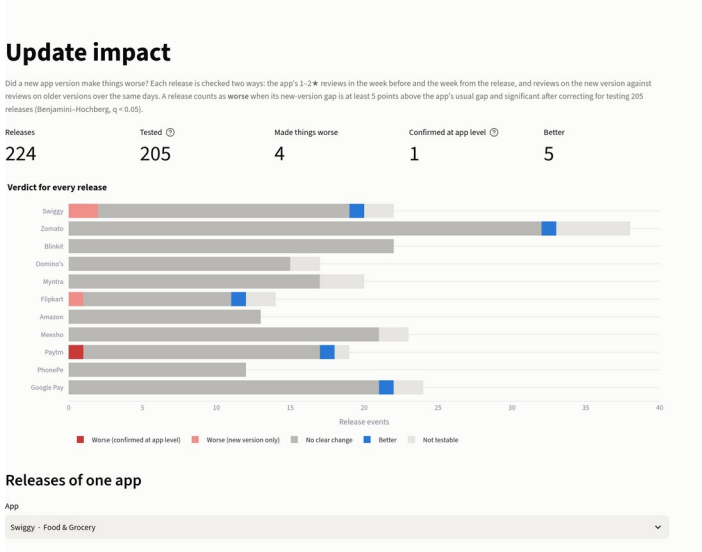
- Investigate 10.84.5 first: 34 → 66 a day
- Plan on last week's level, not the forecast
- Human check on flags (precision 66%)

# Dashboard: Forecasting, Update Impact and Recommendations



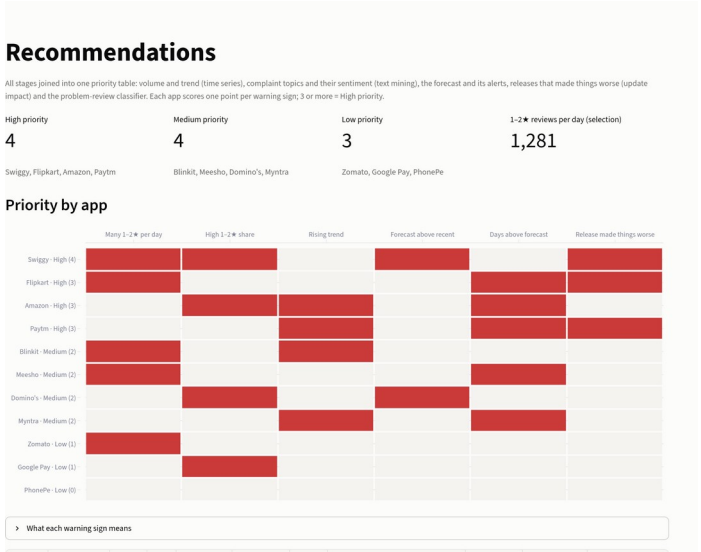

## Forecasting

ARIMA forecast with 80% and 95% intervals per app, unusual days, release markers (red = made things worse), optional baseline comparison




## Update impact

verdict for every release, new vs older gap against the app's usual gap, before/after detail and issue mix of any release




## Recommendations

priority counts, warning-sign grid, priority table and the 65 recommendations, filtered by domain and app, with CSV download

# Project Tracking and Individual Contributions

**Board:** github.com/users/Adithya-Monish-Kumar-K/projects/4 **Repository:** github.com/Adithya-Monish-Kumar-K/ba-capstone-app-reviews

| Backlog | To Do | In Progress | Review/Testing | Completed: 14 of 14 tasks |
|---|---|---|---|---|

| Member | Stage | Board tasks (all due 11 Oct, all Completed) | Pull requests (merged) | Evidence |
|---|---|---|---|---|
| **Aditya Monish Kumar K - CB.SC.U4CSE23103** | 1 and 6 | #131 time series and patterns, #132 text preprocessing, #133 dashboard skeleton and overview; Stage 6: deck, final report, contribution summary | #147, #148, #149 (8 Oct) | scripts 15–16, dashboard/, FINAL_REPORT.md |
| **Regella Krishna Saketh - CB.SC.U4CSE23649** | 2 | #134 topic modelling and distinctive terms, #136 topic evaluation and interpretation | #150, #151 (8 Oct) | scripts 17–18, models/textmining/, TEXT_MINING.md §2–3 |
| **Akshay KS - CB.SC.U4CSE23104** | 3 | #135 ARIMA forecasting with rolling backtest, #139 text-mining dashboard page | #152, #153 (9 Oct) | scripts/19_forecasting.py, 20_topic_examples.py, TIME_SERIES.md §2, text_mining.py |
| **Harshini Vennela - CB.SC.U4CSE23455** | 4 | #137 forecast evaluation and alerts, #142 forecast write-up, #146 check-a-review page | #154, #155, #156 (10 Oct) | scripts/20_forecast_evaluation.py, TIME_SERIES.md §2.5, check_review.py |
| **Kanishka D - CB.SC.U4CSE23155** | 5 | #138 update impact, #141 combined insights, #140 three dashboard pages, #143 final notebook | #157, #158 (11 Oct) | scripts 21–22, COMBINED_INSIGHTS.md, notebooks/Review_2.ipynb |

**Activity history:** one stage after another, each merged before the next began (Stages 1–2 on 8 Oct, Stage 3 on 9 Oct, Stage 4 on 10 Oct, Stage 5 on 11 Oct). Each task is linked to a merged pull request, and every card is in Completed.

# Conclusions, Limitations and Submission Links

## What we found

- Complaints are mostly about service, not the app: refunds, returns, failed transfers and support lead every domain.
- Topic models cover the 43% of complaints that keyword rules missed.
- ARIMA forecasts 1–2★ reviews better than the baseline for 10 of 11 apps, with honest intervals.
- Most releases change nothing; Paytm 10.84.5 clearly made things worse.
- Fix first: Swiggy, Flipkart, Amazon and Paytm.

## Limitations

- 173 days of data: no yearly pattern, only 9 backtest origins.
- A 1–2★ review is a user's judgement, not a confirmed defect.
- VADER underrates calm technical complaints; topic names are our reading of the top terms.
- Users choose when to update, and three of the worse releases overlap the feed gap.
- Counts follow app traffic and campaigns that the models do not see.

## Submission links

**Live dashboard**
team10-app-reviews.streamlit.app

**Final notebook**
notebooks/Review_2.ipynb

**Final report**
FINAL_REPORT.md

**Contribution summary**
docs/Review_2_Contribution_Summary.md

**Project board**
github.com/users/.../projects/4

**Repository**
ba-capstone-app-reviews